Painting Language with Brushstrokes of Mathematics

The Magical Journey of Non-linear Mathematics

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Playing with the mapping of mathematics and language

> Some of you will know that I paint meaning into geometrical space. Is this a trick? “Words in space” sounds like a metaphor, and it is, yet it is also literal. Read on for the magic, and let the meaning flow. If you’re new to these ideas, it may take a few sentences, or even paragraphs, for the shapes to come into view, but they will. Keep going; I’m sure you’ll find the meaning is worth your attention.
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> In that geometric space, language is not a static code. It is a tide carving patterns in the sand, a train tracing unseen rails across a manifold of meaning. And importantly this is more than metaphor: mathematics, language, and even hallucinations move along parallel tracks. In my latest work, I bring mathematics and metaphor into the same phase space and map how words become attractors, how prompts act as symmetry-breaking fields, and how readers complete the loop of meaning.
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> This isn’t just a theory of large language models; it’s an exploration of language itself as a non-linear dynamical system. And this is a finite dance we can chart, perturb, and refine.
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> Different languages, and even mathematics, follow similar curves each with its own curves of meaning, a manifold, that unfolds as we read, the sentences fitting into our own inner landscape. The mathematical map is simply another language, one you may not speak: and that’s fine. I can’t speak Chinese, but I can appreciate the beauty of its characters. The paper below is therefore as much art as mathematics: constituting equations as visual metaphors, symbols as words, and both carrying curvature in a shared dance through space.

The full paper, with equations, detailed framework, the metaphor and the art is available as a PDF here:

→ **The Mathematical Art of Language: The Magical Journey of Nonlinear Mathematics in LLMs**